Unit 5: Thermodynamics, Radiation, Oscillations and Cosmology

IA2 compulsory unit

Externally assessed

5.1 Unit description

Introduction	This topic covers thermal energy, nuclear decay, oscillations and astrophysics and cosmology. This topic may be studied using applications that relate to thermodynamics, for example space technology, and to nuclear radiation, for example nuclear power stations and medical physics. This topic also enables students to develop practical and mathematical skills.
Practical skills	In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include investigating the relationship between the volume and temperature of a fixed mass of gas, measuring the half-life of a radioactive material, measuring gravitational field strength using a simple pendulum and measuring a spring constant from simple harmonic motion.
Mathematical skills	Mathematical skills that could be developed in this topic include substituting numerical values into algebraic equations using appropriate units for physical quantities, applying the concepts underlying calculus (but without requiring the explicit use of derivatives or integrals) by solving equations involving rates of change, for example. $\Delta x/\Delta t = -\lambda x$ using a graphical method or spreadsheet modelling and understanding probability in the context of radioactive decay, sketching relationships that are modelled by $y = \sin x$, $y = \cos x$, $y = k/x$, $y = k/x^2$